Restaurant POS & QR Ordering System Proposal

1. Executive Summary

This proposal presents the development of a Restaurant POS and QR Ordering System from scratch using NestJS, React/Next.js, PostgreSQL, TypeORM, Redis, Socket.IO and Docker. The system will digitize restaurant operations through QR ordering, real-time communication, billing, analytics and centralized administration. The project will be delivered in 90 working days.

2. Understanding of Requirements

The solution will support Customer, Kitchen Staff, Waiter and Administrator roles. It will include QR ordering without login, menu management, table management, billing, payment QR, real-time notifications, analytics, reporting, invoices and responsive interfaces exactly as described in the client's requirements.

3. Proposed Solution

The application will consist of Customer, Kitchen, Waiter and Admin interfaces connected through a centralized NestJS backend with PostgreSQL. WebSockets will synchronize order updates instantly while Redis manages notifications and caching.

4. Technology Stack

Backend: NestJS
ORM: TypeORM
Database: PostgreSQL
Realtime: Socket.IO + Redis
Frontend: React / Next.js
Styling: Tailwind CSS
Deployment: Docker on Cloud VPS
Security: HTTPS, JWT Authentication, Role-Based Access Control

5. Strategic Advantages

• Built entirely from scratch.
• Modular and scalable architecture.
• Multi-branch ready.
• Real-time communication.
• Industry-standard technologies.
• Full ownership transferred after final payment.

6. Delivery Methodology and Timeline

Phase	Duration	Activities
Requirement Analysis & Planning	Days 1–10	Requirements, architecture, database, wireframes,

		project setup
UI/UX Design	Days 11–20	Responsive interface design and approval
Backend Development	Days 21–50	NestJS APIs, authentication, PostgreSQL, QR ordering, billing, analytics, WebSockets
Frontend Development	Days 51–75	React/Next.js interfaces, API integration, responsive UI
Testing & QA	Days 76–85	Testing, optimization, UAT, bug fixing
Deployment & Handover	Days 86–90	Deployment, training, documentation, go-live support

7. Cost and Payment Structure

Total Development Cost: NPR 415,000 (Fixed).

30% Advance
30% After Development Milestone
40% Upon Final Deployment.
Hosting, domain and third-party messaging charges excluded.

8. Ownership and Commercial Terms

Full ownership of the deployed application, source code, database, documentation, branding and configuration transfers to the client after final payment.

9. Risk Mitigation

Structured planning, version control, secure authentication, encrypted communication, regular testing, backups and documentation reduce project risks.

10. Post-Delivery Support

Three months of post-deployment support is recommended. Extended maintenance can be arranged separately.

11. Conclusion

This proposal delivers a secure, scalable and future-ready Restaurant POS system developed from scratch using modern technologies within 90 working days at a fixed cost of NPR 415,000.